

TABLE OF CONTENTS

M 007 VERTICAL CENTRIFUGAL PUMPS

	Page No.
1 SCOPE OF WORK	M 007-2
2 CODES AND STANDARDS	M 007-2
3 DESIGN AND OPERATING CONDITIONS.....	M 007-3
4 DETAILED REQUIREMENTS	M 007-3
4.1 General.....	M 007-3
4.2 Pump Type and Design.....	M 007-3
4.3 Pump Characteristic	M 007-5
4.4 Shaft and Impellers	M 007-5
4.5 Wear rings, Shaft Sleeves, Seals	M 007-6
4.6 Bearings	M 007-6
4.7 Lubrication	M 007-6
4.8 Pump Support Base.....	M 007-6
4.9 Painting.....	M 007-7
4.10 Noise	M 007-7
5 PERFORMANCE TESTING AND GUARANTEES	M 007-7
5.1 Shop Test	M 007-7
5.2 Site Test	M 007-7
6 QUALITY ASSURANCE /QUALITY CONTROL (QA/QC).....	M 007-7
7 INSTRUMENTATION AND CONTROLS	M 007-8
8 ELECTRICAL SPECIFICATIONS.....	M 007-8
8.1 General.....	M 007-8
8.2 Motors.....	M 007-8

M 007 VERTICAL CENTRIFUGAL PUMPS

1 SCOPE OF WORK

The vertical centrifugal pumps specified in **Part 4, Subsection 1.2 "Summary of Work", in Section 3 Technical Specifications** or as required by the Bidder's design and the required electrical equipment specified in **Clause 8** herein, shall be subject to the design, testing and quality assurance requirements of this Specification as applicable.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 "Codes & Standards"**. The Codes and Standards of the following organizations extracted from **Subsection 2.5** or international equivalent standards are specifically applicable to the design manufacture and testing of the equipment included in this specification.

AISI	American Iron and Steel Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
AWWA	American Water Works Association
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standards
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
JIS	Japan Industrial Standard

3 DESIGN AND OPERATING CONDITIONS

A minimum scope of design and operating data shall be submitted by the Bidder for each vertical centrifugal pump as requested in **Schedule 7 “Technical Data”**.

4 DETAILED REQUIREMENTS

4.1 General

Each pump and its appurtenances and accessories shall conform to the requirements of this Specification.

The Total Head (H) specified by the Bidder to the pump manufacturer shall be calculated based on commercial pipes (not new and clean)

4.2 Pump Type and Design

Pumps shall be of the single or multiple stage, vertical centrifugal or mixed flow submerged type for wet-pit service.

The pump operating speed shall not exceed 1500 rpm.

Driving motor shall be mounted on a substantial pump support base mounted on the pump supporting floor with the pump bowl and impeller assembly suspended into the water intake chamber below by means of a column pipe. Bowl assembly shall be flanged construction.

The column pipe shall serve to conduct the water from the impeller upward to the pump discharge outlet.

Pump suction housing shall be long enough to suit the conditions and shall have a belied or flared inlet.

Pump discharge outlet shall be flanged. Discharge flange shall match the ANSI or AWWA Standard pipe flange.

Pump impellers shall be connected to the rotor through an intermediate or line shaft and a suitable keyed coupling. Coupling shall have provisions for making axial adjustments for impeller clearance. Line shaft shall be supported intermittently throughout its length by means of guide bearings. For pumps handling corrosive or contaminated liquid, the line shaft shall be enclosed in a

shaft-enclosing tube for guide bearing support to facilitate proper bearing lubrication, and to protect shaft and guide bearings from contaminated liquid.

Pumps shall be of the heavy-duty type suitable for continuous or intermittent service under all operating conditions as specified, and shall operate without undue strain or wear and without damage to any part of the pumping unit. It shall be Bidder's responsibility to determine the exact type of pump design which will provide such service without operating difficulty or undue replacement of parts.

Pumps shall be designed and constructed to withstand the most severe service conditions under which the pumps will be required to operate. All materials of construction shall be identified with ASTM or AISI standard numbers. All features of design and construction, and all materials, shall be acceptable to the Owner.

Driving motor in each case shall be of the full, direct-across-the-line voltage starting type, and all parts of the pump unit shall be designed to withstand the resulting stresses. Pump and motor design shall be such that no damage will result from reverse rotation, including reverse rotation at startup due to discharge valve failure in discharge line.

Bidder shall be responsible for determining that the pump and motor operate in dynamic balance as a unit without undue vibration or noise throughout the range of operation. Bidder shall verify that the critical speed of the complete rotating unit (motor and pump) is at least 25% above the operating speed.

Bidder shall furnish all such facilities required for proper operation and testing of the pump. Bidder shall perform flow model tests of wet- pit inlet structure to determine whether baffles or other special provisions are required to ensure proper pump performance. The tests will be witnessed by the Owner. Bidder shall notify the Owner within three weeks prior to model testing.

Shop Assembly: Each pump shall be preassembled in the Bidder's plant prior to shipment to ensure proper fit of all components. Each subassembly requiring field assembly shall be match marked and provided with a positive alignment device (fit-up lugs, rabbetted fits, etc.) to facilitate field erection and alignment. The parts of the pump shall be piece marked (by number) with markings corresponding to those on Bidder's drawings.

Each pump shall be arranged so that the entire pump unit may be removed from the support floor for maintenance and repair without dismantling the discharge pipe.

Lifting lugs or eyes shall be provided to facilitate handling.

4.3 Pump Characteristic

Pumps shall have non-overloading characteristics, and shall have stable head-flow characteristics similar to that of a horizontal centrifugal pump so that the head shall continue to increase with a decreasing flow until a maximum head is reached at zero capacity. The design head-flow characteristics shall be acceptable to the Owner.

Duplicate pumps designed for the same flow and head conditions shall operate satisfactorily singly, or in parallel with each other, or in parallel with other pumps of similar characteristics.

Each pump shall operate satisfactorily without distress in any part through full capacity range from zero to maximum specified flow.

The pump length and hydraulic design shall be as follows:

- The pump characteristics shall include net positive suction head required (NPSHR) curves, the recommended minimum NPSHR for normal continuous operation shall be indicated on the curves, for all flows including maximum run out
- Length of the pump shall be established by the Bidder on the basis of the minimum liquid elevation and continuous operation at flows ranging from minimum flow to run-out.
- Length of the pump shall provide sufficient submergence above the pump inlet to prevent vortexing and shall provide the NPSH required.

Before manufacture of a pump rated larger than 150 kW is started, Bidder shall submit to the Owner, the design pump head-flow characteristics.

4.4 Shaft and Impellers

Fastening with impeller nuts is prohibited. Pump shaft shall be supported by radial bearings. Impellers shall be statically and hydraulically balanced radially to insure smooth operation under all conditions.

4.5 Wear rings, Shaft Sleeves, Seals

Impeller and casing or bowl wear rings shall be provided. All wear rings shall be mechanically fastened with a tongue and groove device and locked in place.

Stuffing boxes shall be extra deep with highest quality packing. Packing glands shall be of the split ring type accessible for easy adjustment or replacement of packing without disassembly of pump.

If mechanical seals are used a spacer type coupling shall be provided to allow removal of the seal without removing the driving motor.

Shaft sleeves shall be provided at the stuffing boxes and expected wear points to protect the shafts. Surfaces in contact with packing shall be ground smooth

4.6 Bearings

Bearings shall be protected from abrasive materials that may be present in the water being pumped.

Thrust bearings shall be of the Kingsbury or Michell type and shall be located with the driving motor, and mounted in an accessible location.

4.7 Lubrication

The pumps shall be lubricated by the fluid being pumped. No external lubrication shall be required. Equipment shall be provided to remove impurities from the fluid wherever necessary.

4.8 Pump Support Base

A heavy cast iron or welded steel base shall be furnished for mounting the motor drive and for supporting the pump column and bowl assembly. Base mounting flange shall have machined surface on the underside and shall have provisions for mounting on the curb ring or frame specified below.

The support base shall incorporate the pump discharge elbow, if the pump is required to discharge above the support floor. The base shall be provided with large openings in the sides for easy access to shaft coupling, upper bearing, stuffing box, etc.

The motor support shall be designed to avoid vibrations associated with natural frequency harmonics. The natural frequency of the motor and the motor support on the pump shall be at least 25% above or below the operating speed.

If the high water level is above the support floor the curb ring or frame shall be of watertight construction and furnished with seal plate, gaskets, and bolting, as required.

4.9 Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and all exterior ferrous surfaces shall be painted in accordance with the requirement in **Part 4 Section 3, S-420 and S-430**.

4.10 Noise

The Bidder shall meet the requirements specification **Part 4, Subsection 2.9**.

5 PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Test

All vertical centrifugal pumps shall be shop tested in accordance with Hydraulic Institute Standards, unless the Bidder can show that similar size and type of pump have been tested and the test results are applicable to the pump offered.

5.2 Site Test

After equipment is into put service, Bidder shall run a check test on the equipment to ascertain that it is performing properly in every respect. Testing shall be in accordance with the, guidelines and requirements contained in the "Hydraulic Institute Standards". Any deficiencies brought out by this test shall be promptly corrected by Bidder at no cost to Owner.

6 QUALITY ASSURANCE /QUALITY CONTROL (QA/QC)

Quality Assurance shall conform to the requirements of **Part 4, Subsection 2.7**.

7 INSTRUMENTATION AND CONTROLS

Technical requirements governing the design and supply of instrumentation and control system and devices shall be as detailed in **Part 4, Section 3, I-001**.

The Bidder shall furnish a control panel for each pump or group of pumps where required for normal operation, maintenance, safety and testing purposes. Each panel shall contain all necessary control switches, indicator lights, meters, annunciator control system hardware/modules etc.

Refer to **Part 4 Section 3, I-002, Local Control Panels** for the design requirements for the Pumps Local Control Panels.

8 ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for Vertical Centrifugal Pumps.

For general electrical requirements, see **Part 4, Section 3, E-001**.

8.2 Motors

For detailed electrical requirements, see **Part 4, Section 3, E-210** for low voltage motors.

(Last page of M-007)